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Rocket Engine Analysis

Extension of Rocket Engines – Final Assignment



SE-21 D

SSME RS-25

Group 2

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[1] masamic. *Second Stage Engine of H-I Rocket (LE-5)*. *Wikimedia Commons*, 19 May 2007, commons.wikimedia.org/wiki/File:LE-5.JPG. Licensed under Creative Commons Attribution-ShareAlike 3.0 Unported (CC BY-SA 3.0).

[2] NASA. *Shuttle Main Engine Test Firing*. *Wikimedia Commons*, 1 Jan. 1981 or 21 May 1981, commons.wikimedia.org/wiki/File:Shuttle_Main_Engine_Test_Firing.jpg. Public domain image

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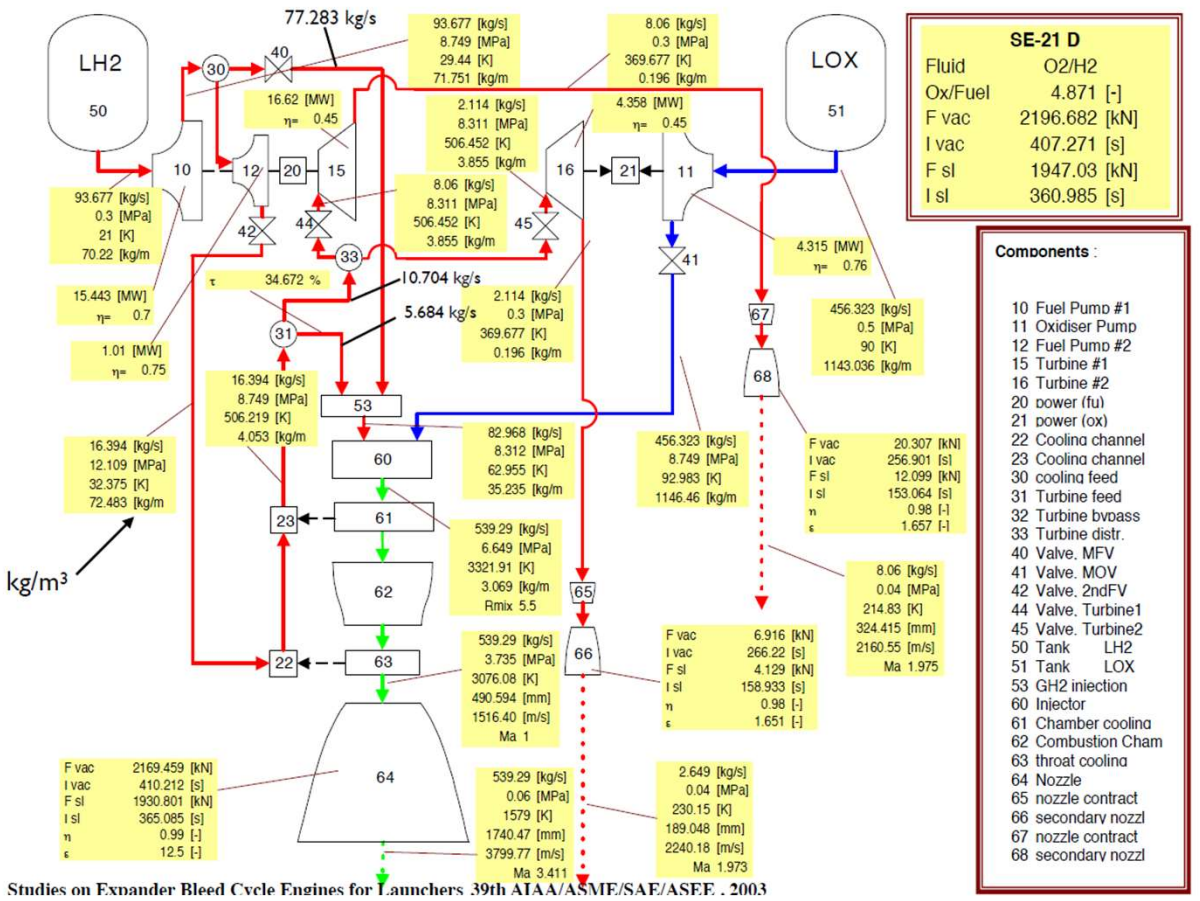
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- Overview of the Engines
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Introduction: SE-21 D

- Expander bleed cycle engine
- Open cycle → turbines powered by gas expansion, hot gas exhausted afterwards
- Development and test designations
- Primarily evaluated for integration into reusable booster systems



[3] Sippel, Martin, et al. "Studies on expander bleed cycle engines for launchers." *39th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit*. 2003.

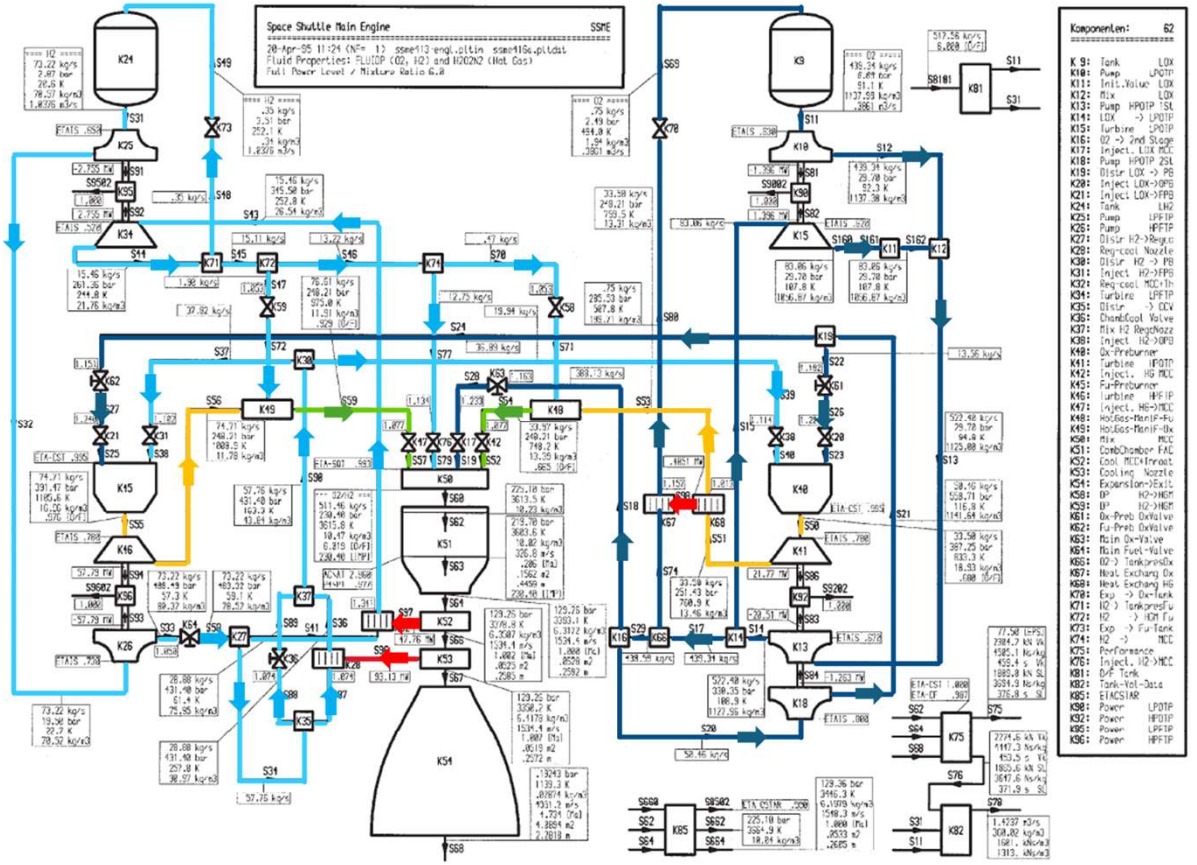
[4] Thakur, Bhaskar, and Indra Jyoti Pegu. "A review of cryogenic rocket engine." *International Research Journal of Engineering and Technology (IRJET)* 4.08 (2017): 2395-0056.

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Introduction: SSME RS-25

H_2 | O_2 | Mix Combu | H_2 + mix | heating

- Space shuttle main engine
- Cryogenic liquid-fuel rocket engine
- Closed cycle, staged combustion
- Reusable around 20 times



[5] Herbertz, Armin. "Component modeling for rocket engine cycle analysis." *Transactions of the Japan Society for Aeronautical and Space Sciences, Aerospace Technology Japan* 14.ists30 (2016): Pa_119-Pa_127.
[6] Bradley, Douglas, and Katherine Van Hooser. "Space shuttle main engine-The relentless pursuit of improvement." *AIAA space 2011 conference & exposition*. 2011.

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Analysis

- Modelling each component of the engine with a function in Matlab:

1. Get the inputs for each stage:
   a. Inlet properties (T,p,ρ,$\dot{m}$,...)
   b. Component chacteristic
   c. Work performed/taken
2. Getting as outputs the outlet flow properties
3. Each stage validated individually
4. Engine validation with error propagation

- Use of HGS or NFP through all the analysis

Except for the heat exchangers

NFP
- Used for single species flows
- Used for cryogenic fluids
- Usability is checked every time

HGS
- Used for gas mixtures
- Functions use HGS if molar amount vector is detected
- Total properties need to be converted
- No direct density detection → find Rg and use ideal gas equation

Checking Fluid State

Determination of thermodynamic regime

Inputs:
- Pressure
- Temperature
- Species
- For mixtures: molar vector

Comparing to critical pressure, critical temperature, saturation temperature

Outputs:

NFP
- Liquid
- Two-phase
- Compressed liquid
- Supercritical
- Gaseous

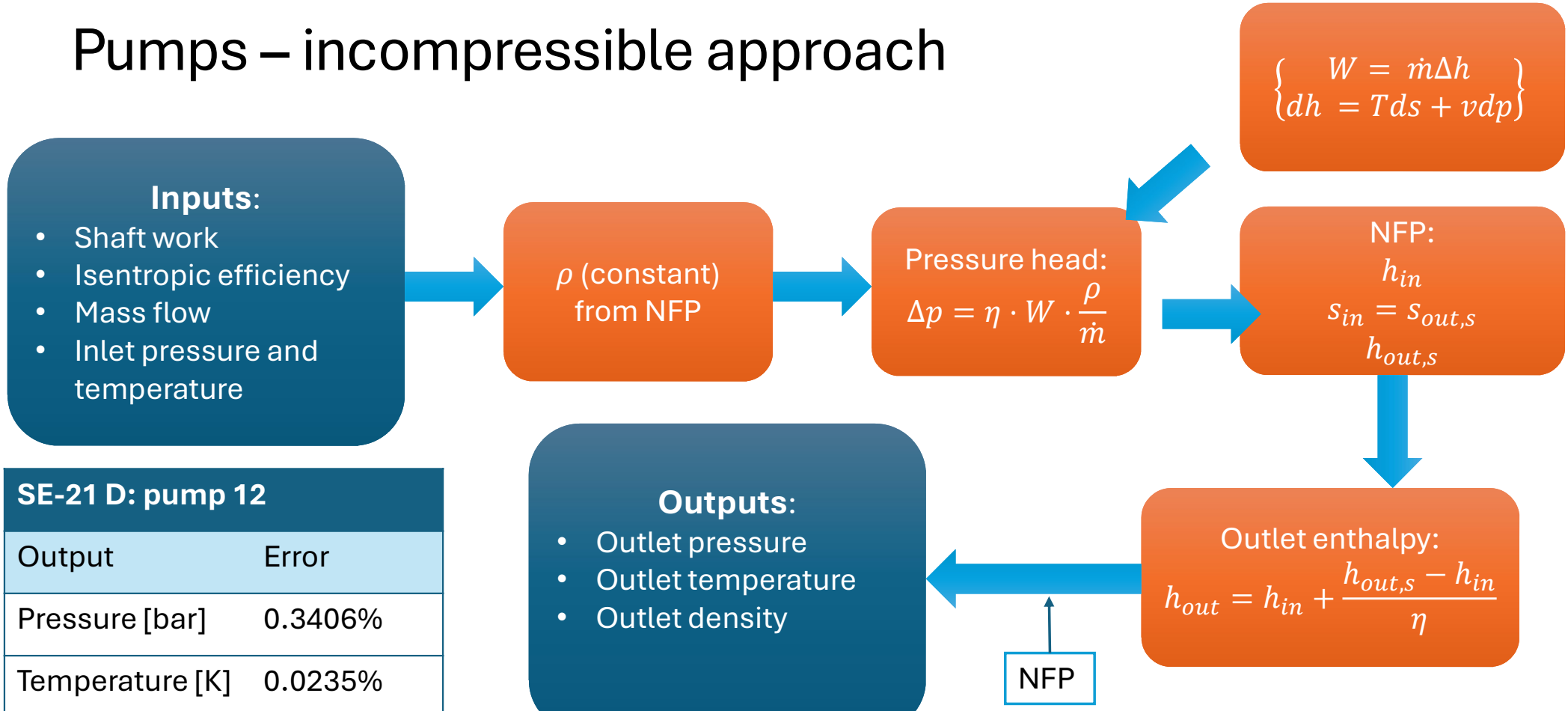
HGS
- High-temperature ideal gas
- Low temperature, validity questionable

Pumps

- Used to transport fuel and oxidizer from tanks into feeding system
- Powered by turbines
- Only for single species fluids → always uses NFP
- Two approaches
  - First: assuming constant density
  - Second: assuming compressible flow

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Pumps – incompressible approach



SE-21 D: pump 12	
Output	Error
Pressure [bar]	0.3406%
Temperature [K]	0.0235%
Density [kg/m^3]	0.1143%

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Pumps – compressible approach

Inputs:
- Shaft work
- Isentropic efficiency
- Mass flow
- Inlet pressure and temperature

NFP:
h_{in}
$s_{in} = s_{out,s}$
$h_{out,s}$

Residual function:
$$f = \frac{1}{\eta} \cdot \int_{p_{in}}^{p_{out}} \frac{1}{\rho} dp - \frac{W}{\dot{m}}$$

```
function val = integrand(p, species, s_in)
    pbar = p/1e5;                          % Pa → bar
    T    = NFP(species,'t_ps',pbar,s_in); % T(p,s)
    rho  = NFP(species,'r_pt',pbar,T);    % ρ(p,T)
    val  = 1./rho;                         % specific
end
```

```
% Residual function for outlet pressure (Pa)
f = @(p_out) ...
    (1/eta) * integral( ...
        @(p) integrand(p, species, s_in), ...
        p_in, p_out, 'ArrayValued', true) ...
    - W/mdot;

% Solve for outlet pressure
p_out = fzero(f, p_guess);
```

Solve by using NFP for integrand function and initial guess for p_{out}

Outlet enthalpy:
$$h_{out} = h_{in} + \frac{h_{out,s} - h_{in}}{\eta}$$

NFP

Outputs:
- Outlet pressure
- Outlet temperature
- Outlet density

Pumps – increase of results

- Example: Pump K26
- Large errors with incompressible assumption → *questionable assumption for rocket engines*
- Errors reduced by up to two orders of magnitude
- Longer run times due to iterative solution

Output	Reference	Incompressible	Compressible
Pressure [bar]	486.49	425.96 -12.4%	486.2 -0.1%
Temperature [K]	57.3	53.712 -6.3%	57.454 +0.3%
Density [kg/m^3]	80.32	70.569 -12.1%	80.043 -0.4%

Turbines

- Generating energy by expanding hot gasses
- Powering the pumps
- Assumption
  - Adiabatic
  - Frozen chemical composition
- Developed both with HGS and NFP

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Turbines – NFP / HGS

Inputs:
- Required Power
- Isentropic efficiency
- Mass flow
- Inlet pressure and temperature

NFP / HGSprop:
$h_{in}\ /\ s_{in} = s_{out,s}$
$H_{in}\ /\ S_{in} = S_{out}$

Turbine power balance:
$h_{out} = h_{in} - \frac{W}{\dot{m}}$
$H_{out} = H_{in} - W$

Isentropic outlet enthalpy:
$h_{out,s} = h_{in} - \frac{h_{in} - h_{out}}{\eta}$
$H_{out,s} = H_{in} - \frac{H_{in} - H_{out}}{\eta}$

Solve set of functions until both are zero
$\begin{Bmatrix} h(p,T) - h_{out,s} = 0 \\ s(p,T) - s_{out,s} = 0 \end{Bmatrix}$

```
% Solve (h(p,T), s(p,T)) simultaneously
fun = @(x) [
    NFP(species,'h_pt',x(1),x(2)) - h_out_s
    NFP(species,'s_pt',x(1),x(2)) - s_out_s
];
x0 = [0.3*pbar_in, 0.8*T_in];   % [p_bar, T]
x_sol = fsolve(fun, x0);

pbar_out_s = x_sol(1);
```

assume
$p_{out} = p_{out,s}$

Outputs:
- Outlet pressure
- Outlet temperature
- Outlet density

NFP / HGSprop

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Secondary Nozzle

- Finding exit conditions using perfect gas rocket equations
  - Steady, one-dimensional, isentropic expansion
  - Gas has perfect gas behaviour
  - Viscous effects, heat transfer, chemical reactions neglected

Known:
- Exit velocity
- Expansion ratio
- Exit Mach number
- Exit diameter
- Thrust
- Mass flow
- Ambient pressure

Gas properties from NFP

γ from isentropic area-Mach relation

$$A_{ratio} = \frac{1}{M} \cdot \frac{2}{\gamma + 1} \cdot \left(1 + \frac{\gamma - 1}{2} \cdot M^2\right)^{\frac{\gamma+1}{2\cdot(\gamma-1)}}$$

Exit pressure from

$$F = \dot{m} * V_e + (p_e - p_a) \cdot A_e$$

Exit temperature

$$T_e = \frac{V_e^2}{Ma^2 \cdot \gamma \cdot R}$$

Outputs
- Exit static pressure and temperature

Slide 13

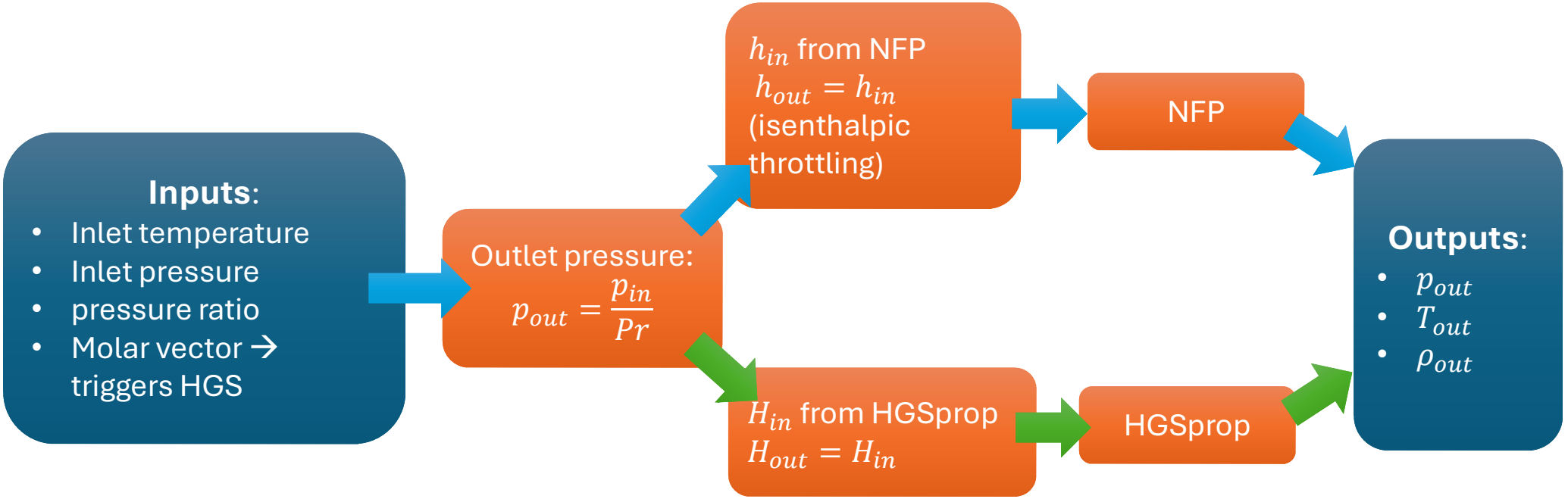
JS1 Maybe redo the function

Jan Sonnenschein; 2026-01-09T12:10:14.194

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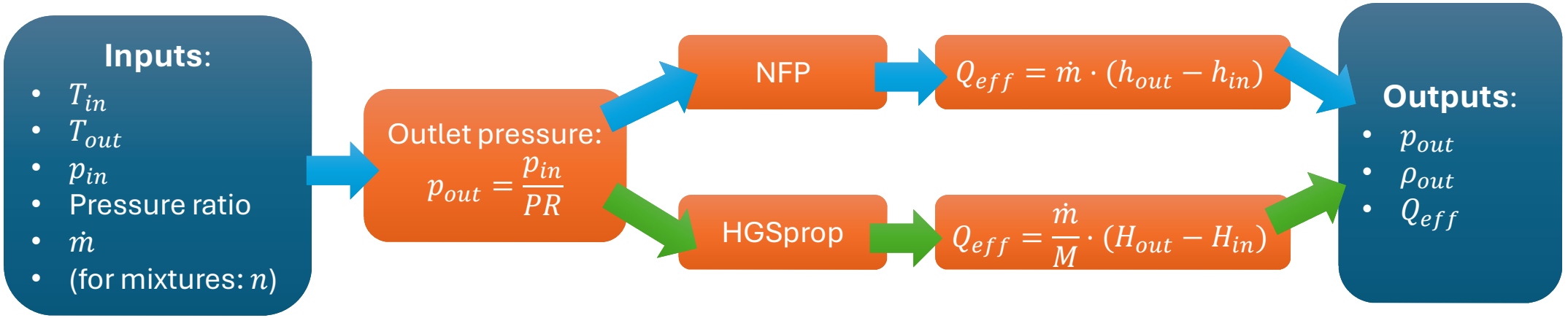
Valves

Steady-state, adiabatic throttling valves



Heat Exchanger

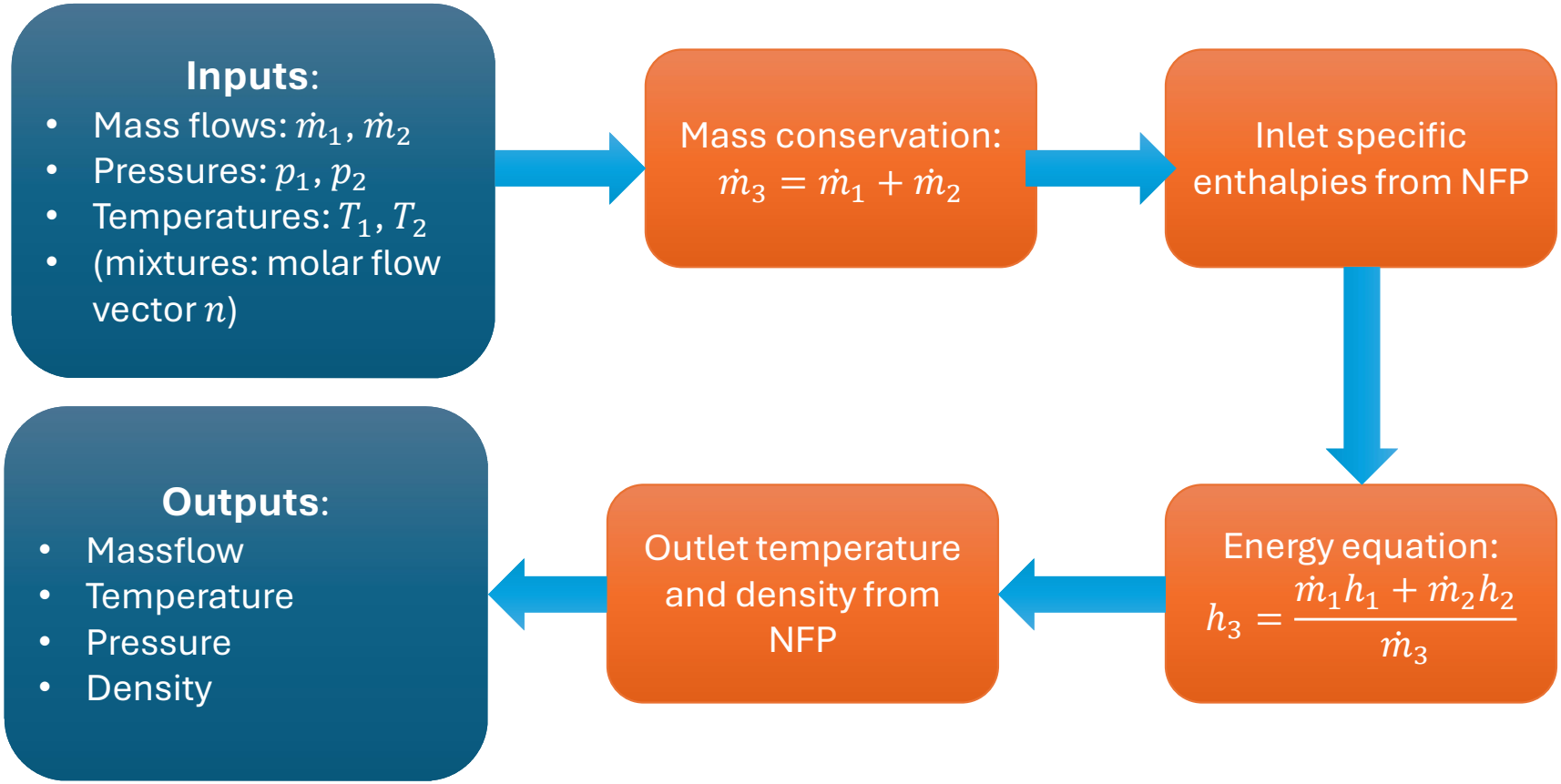
Main goal: finding effective heat transfer rate Q_{eff}



Mixer

- Two streams merge and exit at a **known exit pressure p_3**
- Assumptions
  - Non-reactive
  - Adiabatic: $\dot{m}_1 \cdot h_1 + \dot{m}_2 \cdot h_2 = \dot{m}_3 \cdot h_3$
  - Steady flow
  - No shaft work
  - Negligible kinetic and potential energy

Mixer – NFP


Inputs:
Mass flows: $\dot{m}_1, \dot{m}_2$
Pressures: p_1, p_2
Temperatures: T_1, T_2
(mixtures: molar flow vector n)
Mass conservation: $\dot{m}_3 = \dot{m}_1 + \dot{m}_2$
Inlet specific enthalpies from NFP
Energy equation: $h_3 = \frac{\dot{m}_1 h_1 + \dot{m}_2 h_2}{\dot{m}_3}$
Outlet temperature and density from NFP
Outputs:
Massflow
Temperature
Pressure
Density

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Mixer – HGS

Inputs:
- Mass flows: $\dot{m}_1, \dot{m}_2$
- Pressures: p_1, p_2
- Temperatures: T_1, T_2
- (mixtures: molar flow vector n)

Mass conversation: $\dot{m}_3 = \dot{m}_1 + \dot{m}_2$

Merge species and molar flows

```
species_all = unique([species1, species2],'stable');
n_tot = zeros(1,numel(species_all));

[~,i1] = ismember(species1,species_all);
n_tot(i1) = n_tot(i1) + n1;

[~,i2] = ismember(species2,species_all);
n_tot(i2) = n_tot(i2) + n2;

out.species = species_all;
out.n       = n_tot;
```

Inlet Enthalpies
- For mixtures → HGSprop
- Else → NFP → convert to HGS-reference

Energy balance: $H_3 = H_1 + H_2$

```
% Convert NFP enthalpy to HGS reference (400 K)
delta_h1 = ( ...
    NFP(in1.species,'h_pt',p1_bar,400) - ...
    NFP(in1.species,'h_pt',p1_bar,in1.T) ) * Mm1;
h1 = HGSsingle(in1.species,'h',400);          % [kJ/mol]
H1 = n1 * h1 - delta_h1;                      % [kJ]
```

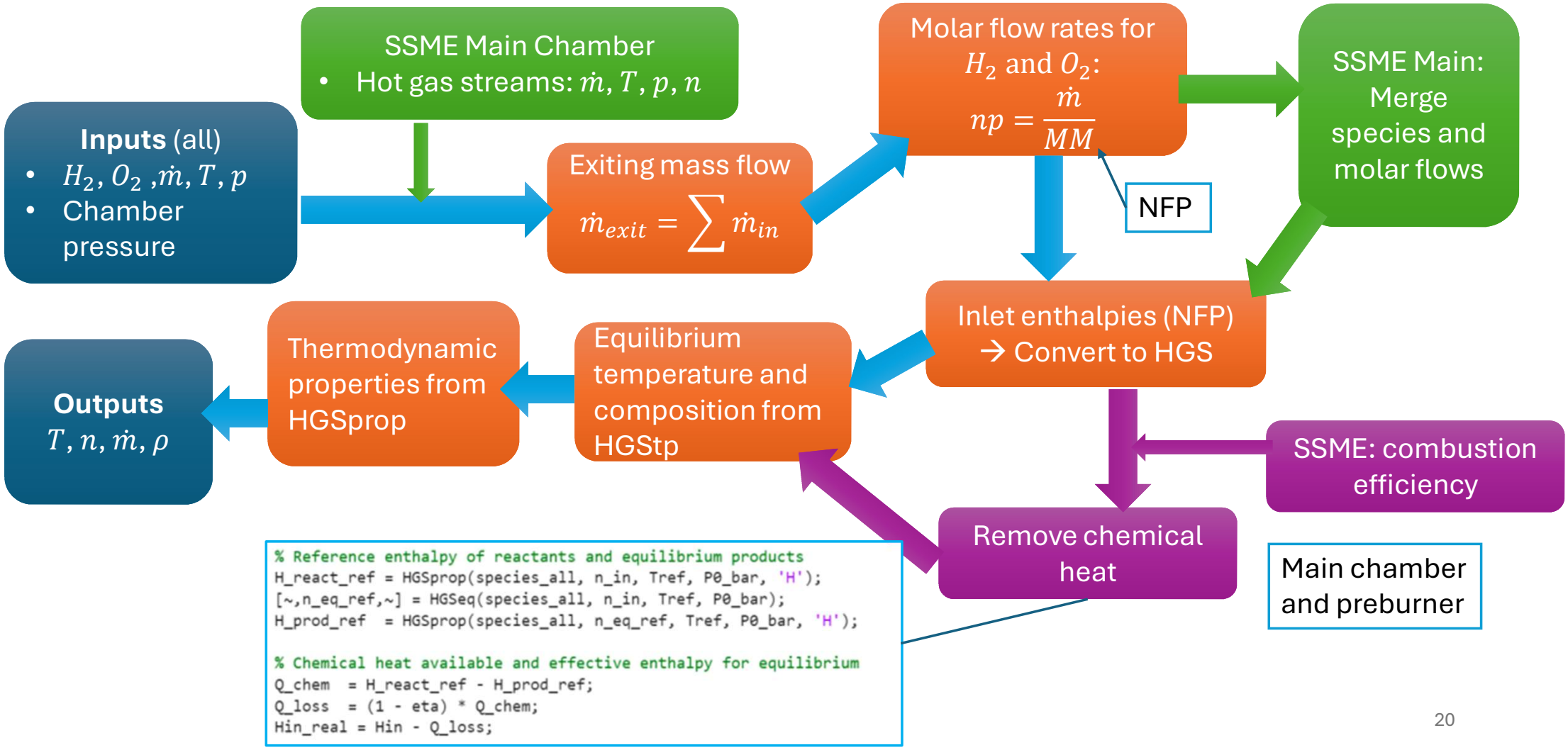
Outlet temperature and density from HGSprop

Outputs:
- Massflow
- Temperature
- Pressure
- Density
- Molar composition

Combustion Chamber

- Different functions for SE_21 D, SSME main chamber and SSME preburners
- Computing equilibrium combustion state
- Using NFP for cryogenic inlet streams
- Using HGS for high-temperature chemical equilibrium at chamber pressure
- Assumptions
  - Steady-state operation
  - Adiabatic combustion chamber
  - Complete mixing prior to reaction
  - Chemical equilibrium at chamber pressure

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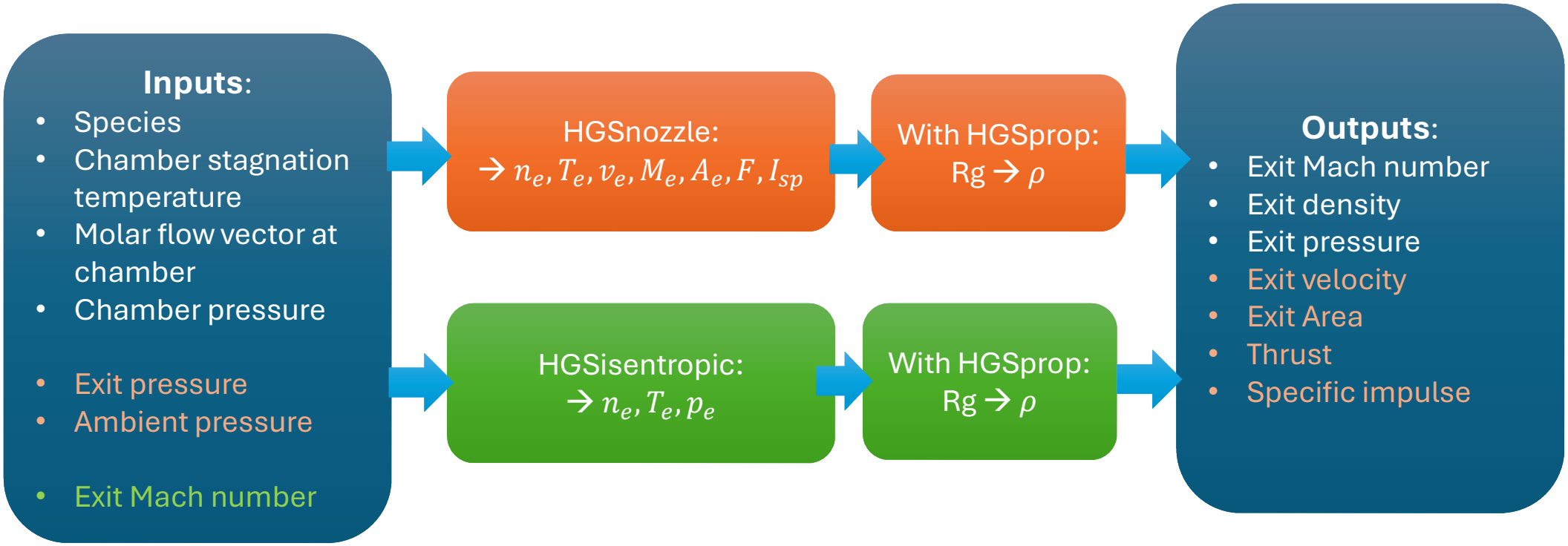

```
% Reference enthalpy of reactants and equilibrium products
H_react_ref = HGSprop(species_all, n_in, Tref, P0_bar, 'H');
[~,n_eq_ref,~] = HGSeq(species_all, n_in, Tref, P0_bar);
H_prod_ref  = HGSprop(species_all, n_eq_ref, Tref, P0_bar, 'H');

% Chemical heat available and effective enthalpy for equilibrium
Q_chem  = H_react_ref - H_prod_ref;
Q_loss  = (1 - eta) * Q_chem;
Hin_real = Hin - Q_loss;
```

Main Nozzle

- First approach: assuming an adiabatic nozzle
- Second approach: nozzle with corrected stagnation conditions
- Using
  - HGSnozzle → full nozzle solution with thrust and Isp
  - HGSisentropic → isentropic expansion to a specified Mach number
- Assumptions
  - Steady-stade
  - One-dimensional
  - Inviscid
- Chemical composition treatment is controlled by Fro_Shift (frozen/ shifting)

Main nozzle – adiabatic



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Main nozzle – corrected

- Previous stagnation enthalpy correction
- Subtracting of heat term Q (computed from heat exchanger or given)
- $\Delta h_0 = -\frac{Q}{\dot{m}}$
- $\Delta H_{expected} = \Delta h_0 \cdot \sum n_p \cdot MM$
- $H_{corr} = H + \Delta H_{expected}$

```
%% Enthalpy correction due to heat extraction
% Q is provided in W and converted to kJ/s for consistency with H
% Hcorr = H - Q/1000;                                % [kJ/s]
delta_h0=-(Q/1000)/mdot; %[kJ/kg]
m_basis = sum(np) * (Mm/1000);  % [kg] if Mm_60 is g/mol
DeltaH_expected = delta_h0 * m_basis;     % [kJ]
Hcorr = H + DeltaH_expected;

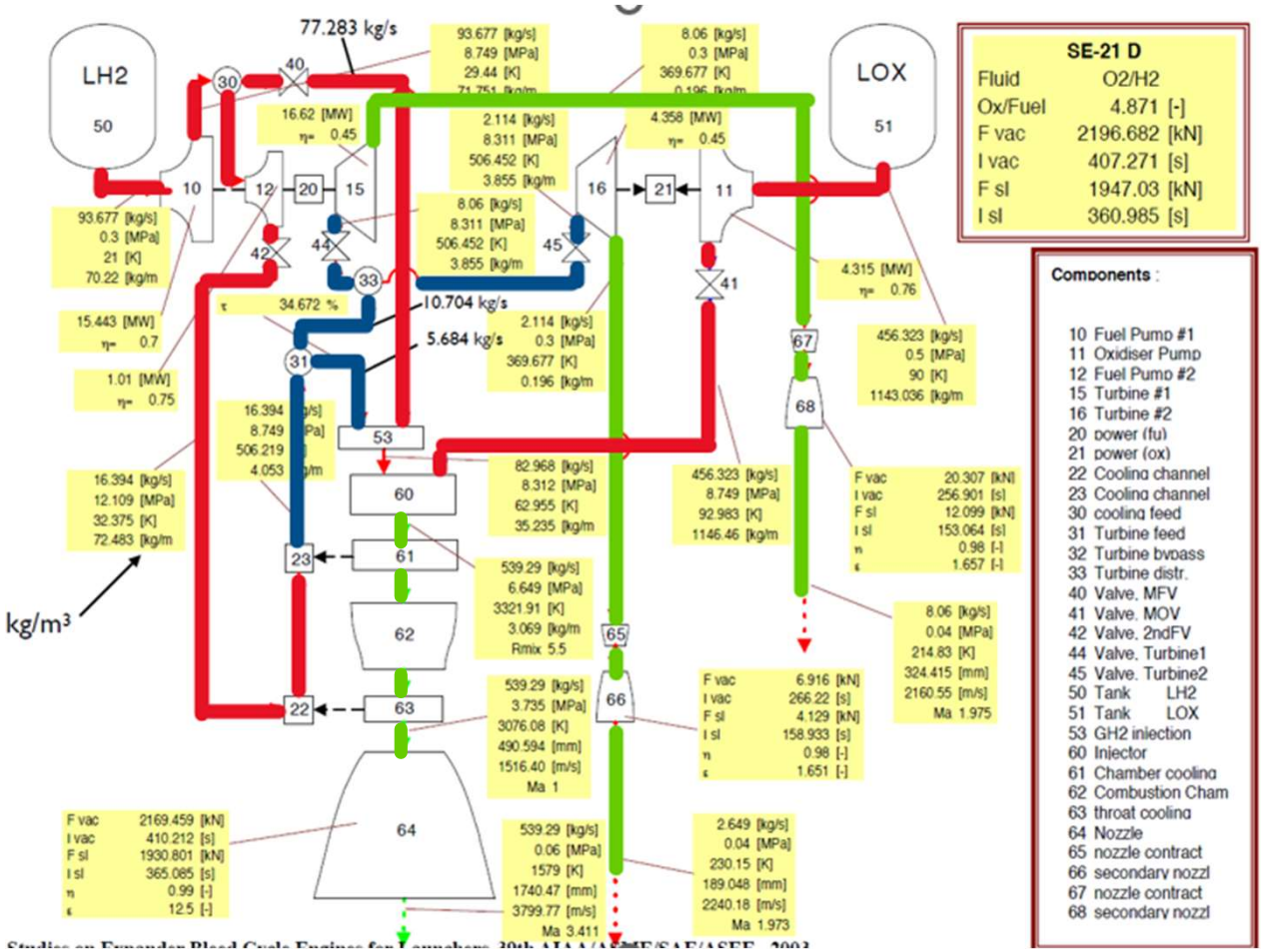
%% Compute corrected stagnation temperature
% The corrected temperature is obtained by inverting the enthalpy
f = @(Tcorr) HGSprop(species,np,Tcorr,Pchamber_bar,'H') - Hcorr;
Tcorr = fzero(f, Tp);
```

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Results – SE-21 D

Fluid state

- Liquid in tanks and through pumps
- Supercritical fluid after cooling section and mixer
- Gas state after pumps
- In nozzle: ideal gas assumed (HGS)



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Results – SE-21 D

Pressure, Temperature, Massflow, Density (selection)

Pumps are very accurate

Position	p (MPa)	p_calc	T [K]	T_calc	rho [kg/m^3]	rho_calc	mdot [kg/s]	mdot_calc
Pump (10)	8.749	8.7528 (0.04 %)	29.44	29.588 (0.01 %)	71.751	71.704 (0.07%)	93.677	93.677
Turbine (15)	0.3	0.28418 (5.27%)	369.677	368.32 (0.38%)	0.196	0.18681 (4.69%)	8.06	8.06
Turbine (16)	0.3	0.77349 (158%)	369.677	396.7 (7.3%)	0.196	0.47101 (140%)	2.114	2.644 (25%)
Turbine (16)	0.3	0.28462 (5.13%)	369.677	368.35 (0.36%)	0.196	0.18707 (4.56%)	2.114	2.114
Secondary Nozzle (66)	0.04	0.037411 (6.9%)	230.15	223.47 (3%)		0.040592	2.649	2.644 (0.2%)

Turbines with 5% error

Biggest problem: Turbine 16
→ Error in reference (wrong mass flow)

Results SE-21 D

- Cooling line (heat exchanger) → $Q_{eff} = 111\ MW$
- Comparison of adiabatic and corrected nozzle
  - Error in exit temperature reduced by 2%
  - Temperature still overestimated
  - Heat loss to environment not considered
  - Friction losses will occur

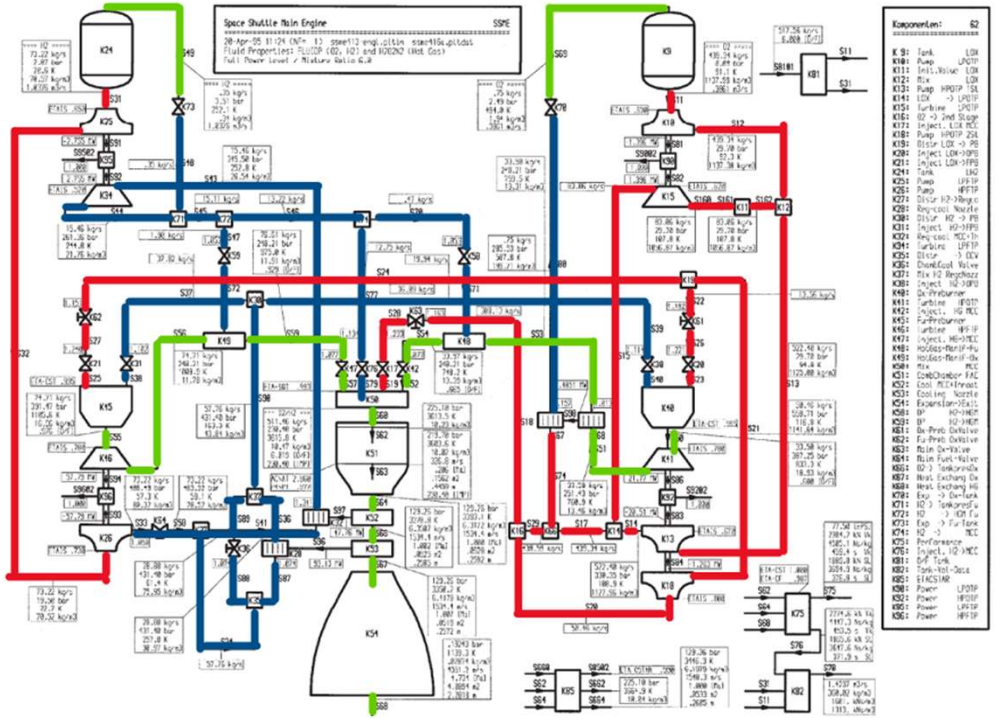
Position	T [K]	T_calc
Nozzle (adiabatic)	1579	1678.3 (6.3%)
Nozzle (corrected)	1579	1647.9 (4.4%)

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Results – SSME

Fluid state

- Hydrogen first in liquid form, supercritical after second pump
- Oxygen stays in liquid form except after heat exchanger
- Mixtures gaseous



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Results – SSME

All functions work nearly perfect

Pressure, Temperature, Massflow, Density (selection)

Position	p (MPa)	p_calc	T [K]	T_calc	rho [kg/m^3]	rho_calc	mdot [kg/s]	mdot_calc
Pump (K26)	48.649	48.653 (0.008%)	57.3	57.367 (0.12%)	80.32	80.095 (0.28%)	73.22	73.22
Mixer (K37)	43.14	43.14	163.3	164.17 (0.53%)	43.84	43.474 (0.83%)	57.76	57.76
Heat Exch. (K32) → Q	34.55	34.55	252.8	252.8	26.54	26.403 (0.52%)	15.46	15.46
Turbine (K15)	2.97	2.9558 (0.48%)	107.8	107.78 (0.02%)	1056.87	1056.9	83.06	83.06
Preburner (K45)	39.147	39.147	1105.6	1112.9 (0.66%)	16.96	16.847 (0.67%)	74.71	74.71
C. Chamber (K50)	23.04	23.04	3615.8	3640 (0.67%)	10.47	10.371 (0.95%)	511.46	511.46

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Results – SSME

Heat exchangers – comparing Q_{eff} with given Q

- Positive heat flows: overestimated → losses to environment
- K67/K68: positive heat flow accurate, losses represented in negative flow

	Q [MW]	Q_{eff} [MW]
K28	93.13	79.57 (14.7%)
K32	47.76	40.58 (15.0%)
K67	0.4051	0.4052
K68	-0.4051	-0.4187 (3.4%)

SSME – Results

Comparing adiabatic with corrected nozzle

- Correction reduces error
- Heat flows given → better results compared to SE-21 D (estimated Q)

Position	T [K]	calc
Nozzle (adiabatic)	1139.3	1196.0 (5.0%)
Nozzle (corrected)	1139.3	1164.4 (2.2%)

→ overall: very good results for SSME
→ better reference data

Results – Error Propagation

- Effects very visible for heat exchangers and combustion chambers
- Heat exchanger K68

- Nozzle temperatures after heat loss correction

K68	Original	Result from function	With error propagation
Effective heat flow [MW]	-0.4051	-0.4187	-2.3734

Engine / Nozzle exit temperature [K]	Original	Result from function	With error propagation
SE-21 D	1579	1647.9	1704.2
SSME	1139.3	1164.4	1177.2

Conclusions

- Reference values for second engine more precise
  - Our models work almost perfectly
- Assuming incompressibility in pumps → highly visible effect in second engine
- Extending functions for challenges occurring at second engine
  - Mixing of cryogenic and hot gases in second engine
  - Efficiency in combustion chamber
- Highest inaccuracies are related to cooling and heat exchanges.
  - Improving error propagation by obtaining actual heat flow going inside the coolant and knowing the pressure losses
- Advantages and disadvantages of using the functions
  - We can see how changing a value effects the results
  - In reality combustion pressure not known

References

[1] masamic. *Second Stage Engine of H-I Rocket (LE-5)*. *Wikimedia Commons*, 19 May 2007, commons.wikimedia.org/wiki/File:LE-5.JPG. Licensed under Creative Commons Attribution-ShareAlike 3.0 Unported (CC BY-SA 3.0).

[2] NASA. *Shuttle Main Engine Test Firing*. *Wikimedia Commons*, 1 Jan. 1981 or 21 May 1981, commons.wikimedia.org/wiki/File:Shuttle_Main_Engine_Test_Firing.jpg. Public domain image

[3] Sippel, Martin, et al. "Studies on expander bleed cycle engines for launchers." *39th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit*. 2003.

[4] Thakur, Bhaskar, and Indra Jyoti Pegu. "A review of cryogenic rocket engine." *International Research Journal of Engineering and Technology (IRJET)* 4.08 (2017): 2395-0056.

[5] Herbertz, Armin. "Component modeling for rocket engine cycle analysis." *Transactions of the Japan Society for Aeronautical and Space Sciences, Aerospace Technology Japan* 14.ists30 (2016): Pa_119-Pa_127.

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